

Q1.

The discrete random variable X has the following probability distribution

x	1	2	4	9
$P(X = x)$	0.2	0.4	0.35	0.05

The continuous random variable Y has the following probability density function

$$f(y) = \begin{cases} \frac{1}{64}y^3 & 0 \leq y \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Given that X and Y are independent, show that $E(X^2 + Y^2) = \frac{1327}{60}$

(Total 4 marks)

Q2.

Participants in a school jumping competition gain a total score for each jump based on the length, L metres, jumped beyond a fixed point and a mark, S , for style.

L may be regarded as a continuous random variable with probability density function

$$f(l) = \begin{cases} wl & 0 \leq l \leq 15 \\ 0 & \text{otherwise} \end{cases}$$

where w is a constant.

S may be regarded as a discrete random variable with probability function

$$P(S = s) = \begin{cases} \frac{1}{15}s & s = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Assume that L and S are independent.

The total score for a participant in this competition, T , is given by $T = L^2 + \frac{1}{2}S$

Show that the expected total score for a participant is $114\frac{1}{3}$

(Total 5 marks)

Q3.

The schedule for an organisation's afternoon meeting is as follows.

Session A (Speaker 1)	2.00 pm to 3.15 pm
Session B (Discussion)	3.15 pm to 3.45 pm

Records show that:

the duration, X , of Session A has mean 68 minutes and standard deviation 10 minutes;

the duration, Y , of Session B has mean 25 minutes and standard deviation 5 minutes;

the duration, Z , of Session C has mean 73 minutes and standard deviation 15 minutes;

and that X , Y and Z are independent.

(a) Determine the means and the variances of:

(i) $L = X + Z$; (2)

(ii) $M = X + Y$. (3)

(b) Assuming that L and M are each normally distributed, determine the probability that:

(i) the total time for the two speaker sessions is less than $2\frac{1}{2}$ hours; (2)

(ii) Session C is late in starting. (3)

(Total 10 marks)

Q4.

X and Y are two independent random variables such that $X \sim N(120, 36)$ and $y \sim N(90, 13)$.

(a) For the random variable $X - Y$, write down the value of:

(i) $E(X - Y)$; (1)

(ii) $\text{Var}(X - Y)$. (1)

(b) Calculate $P(X - Y > 40.5)$. (4)

(c) The random variable T is defined by $T = X_1 + X_2 + 2Y$, where X_1 and X_2 are two independent observations on X .

(i) Find values for the mean and variance of T .

(3)

- (ii) Calculate $P(T < 400)$.

(4)

(Total 13 marks)

Q5.

- (a) Anchal travels to work each day by initially walking from home to the bus stop, waiting for the bus to arrive and then travelling by bus to her office.

The time, A_1 minutes, that she spends walking from home to the bus stop is a normal random variable with a mean of 12 and a variance of 3.

The time, A_2 minutes, that she spends waiting for the bus to arrive is a normal random variable with a mean of 6 and a variance of 2.

The time, A_3 minutes, spent travelling by bus to her office is a normal random variable with a mean of 32 and a variance of 20.

- (i) Assuming that A_1 , A_2 and A_3 are independent, write down the distribution of T_A , the total time, in minutes, for Anchal's journey to work.

(2)

- (ii) Calculate the probability that, on a randomly selected day, Anchal will take less than one hour to travel to work.

(2)

- (b) Baldeep drives to work each day. The total time, T_B minutes, that he spends driving to work is a normal random variable with a mean of 53 and a variance of 16. Calculate the probability that, on a randomly selected day, he will take less than one hour to travel to work.

(1)

- (c) Calculate the probability that, on a randomly selected day, at least one of Anchal and Baldeep will take longer than one hour to travel to work;

(3)

(Total 8 marks)

Q6.

Two independent discrete random variables, M and R , are such that $E(M) = 1.7$ and $\text{Var}(M) = 0.81$, with R having the following probability distribution.

r	2	4	6
$P(R = r)$	$\frac{1}{8}$	$\frac{9}{16}$	$\frac{5}{16}$

- (a) Given that M_1 , M_2 , and M_3 are three independent observations on M , write down the value of:

- (i) $E(M_1 + M_2 + M_3)$;

(1)

(ii) $\text{Var}(M_1 + M_2 + M_3).$

(1)

(b) (i) Calculate values for $E(R)$ and $\text{Var}(R).$

(4)

(ii) Hence find values for the mean and variance of $\frac{1}{2}R.$

(3)

(c) The discrete random variable T is defined by:

$$T = \frac{1}{2}R + M_1 + M_2 + M_3.$$

Evaluate:

(i) $E(T);$

(1)

(ii) $\text{Var}(T).$

(2)

(Total 12 marks)

Q7.

A triathlon comprises three events: a swim, a cycle ride and a run.

The time, S minutes, to complete the swim is a normal random variable with mean 32 and standard deviation 5.

The time, C minutes, to complete the cycle ride is a normal random variable with mean 40 and standard deviation 8.

The time, R minutes, to complete the run is a normal random variable with mean 58 and standard deviation 12.

The variables S , C and R may be considered to be independent.

The random variable X is defined by $X = S + C + R.$

(a) State, in context, what X represents.

(1)

(b) Find:

(i) $E(X);$

(1)

(ii) $\text{Var}(X).$

(2)

- (c) Calculate $P(X > 146)$.

(3)

- (d) The competitors compete in teams of three. Each competitor completes all three events.

Given that each team consists of competitors selected at random, calculate the probability that the total time by a team to complete the triathlon is greater than 412 minutes.

(4)

(Total 11 marks)

Q8.

Each morning, Jack's journey to work consists of four parts: car, transfer, rail and walk. The random variables denoting the durations, in minutes, of the four parts of his journey are C , T , R and W respectively.

The mean and standard deviation of the duration of each part of his journey are shown in the table.

Part of journey	Mean (minutes)	Standard deviation (minutes)
Car	20	3
Transfer	10	3
Rail	75	10
Walk	10	2

The duration of each part of his journey may be assumed to be normally distributed and independent of other parts.

Calculate the probability that, on a particular day, Jack's journey to work takes less than 2 hours.

(Total 5 marks)

Q9.

Cartons of semi-skimmed milk have weights that may be assumed to be normally distributed with a mean of 1085 grams and a standard deviation of 18 grams.

A box holds 6 randomly selected cartons. The weights of these boxes are independent of the weights of the cartons and may be assumed to be normally distributed with a mean of 420 grams and a standard deviation of 6 grams.

- (a) Calculate values for the mean and variance of the weight, W , of a box plus 6 cartons.

(3)

- (b) Hence determine the probability that W is less than 7 kilograms.

(2)

(Total 5 marks)

Q10.

At a particular university, all first year students are required to visit a Registry Desk and a Finance Desk as part of the enrolment process.

Times, X seconds, at the Registry Desk are normally distributed with mean 220 and standard deviation 20.

Times, Y seconds, at the Finance Desk are independent of those at the Registry Desk and are normally distributed with mean 175 and standard deviation 40.

Determine the probability that a first year student spends a **total** of less than 5 minutes at the Registry and Finance Desks;

(Total 5 marks)

Q11.

Bags of carrots have weights that are normally distributed with a mean of 26.8 kg and a standard deviation of 0.9 kg.

- (a) Determine the probability that the **total** weight of a random sample of 25 bags is less than 675 kg.

(5)

- (b) Determine the probability that the difference in weight between two randomly selected bags is more than 1 kg.

(6)

(Total 11 marks)

Q12.

The time taken by a householder to mow the lawns using lawnmower H may be assumed to be normally distributed with a mean of 130 minutes and a standard deviation of 15 minutes. ©2025 Exam Papers Practice. All rights reserved.

The time taken by the householder to mow the same lawns using lawnmower R may also be assumed to be normally distributed, but with a mean of 115 minutes and a standard deviation of 10 minutes.

During a particular year, the householder mows the lawns on 15 occasions using lawnmower H and on 35 occasions using lawnmower R . All mowing times may be assumed to be independent.

Determine the probability that, during this year, the householder's total time in mowing the lawns is more than 100 hours.

(Total 6 marks)